

Cell Biology Structures and Transport

Cells maintain internal conditions through specialized structures and controlled transport. These notes explain organelle functions, membrane transport, and a simple osmosis experiment.

Cell structures

The plasma membrane is a selectively permeable boundary built mainly from a phospholipid bilayer and proteins. It controls movement into and out of a cell.

The nucleus contains most of a eukaryotic cell's DNA. Ribosomes assemble proteins from amino acids using instructions carried by messenger RNA.

The rough endoplasmic reticulum has attached ribosomes and helps process proteins destined for secretion or membranes. The Golgi apparatus modifies and sorts proteins for delivery.

Mitochondria produce much of the ATP used by eukaryotic cells during aerobic respiration. Chloroplasts carry out photosynthesis in plants and algae.

Plant cell walls provide structural support and contain cellulose. A large central vacuole stores water and contributes to turgor pressure in many plant cells.

Movement across membranes

Diffusion is the net movement of particles down their concentration gradient. Simple diffusion crosses the lipid bilayer directly; facilitated diffusion uses membrane proteins. Neither process directly requires ATP.

Osmosis is the net movement of water across a selectively permeable membrane. In the simple experiment below, water moves toward the side with a higher concentration of dissolved solute.

Active transport moves substances against an electrochemical gradient using an energy source. The sodium-potassium pump uses ATP to move three sodium ions out and two potassium ions into the cell per cycle.

Endocytosis brings material into a cell using vesicles. Exocytosis releases material when vesicles fuse with the plasma membrane.

Osmosis experiment

A membrane separates solution A, containing 2 percent sucrose, from solution B, containing 10 percent sucrose. The membrane allows water through but blocks sucrose. Assume equal initial pressure on both sides.

Water initially moves from A toward B. This illustrates osmosis; sucrose does not cross the membrane. Changing the membrane so that sucrose can cross would change the experiment.